

Surge Pricing: The Revenue Engine That's Destroying Retention

62.8% of contribution margin comes from just **31.2% of trips** — but **satisfaction collapses 43%** across the surge range, and **39.6% of all trips lose money** before a single rupee reaches the bottom line.

62.8%
CM from Surge Trips
vs 37.2% from 68.8% non-surge volume

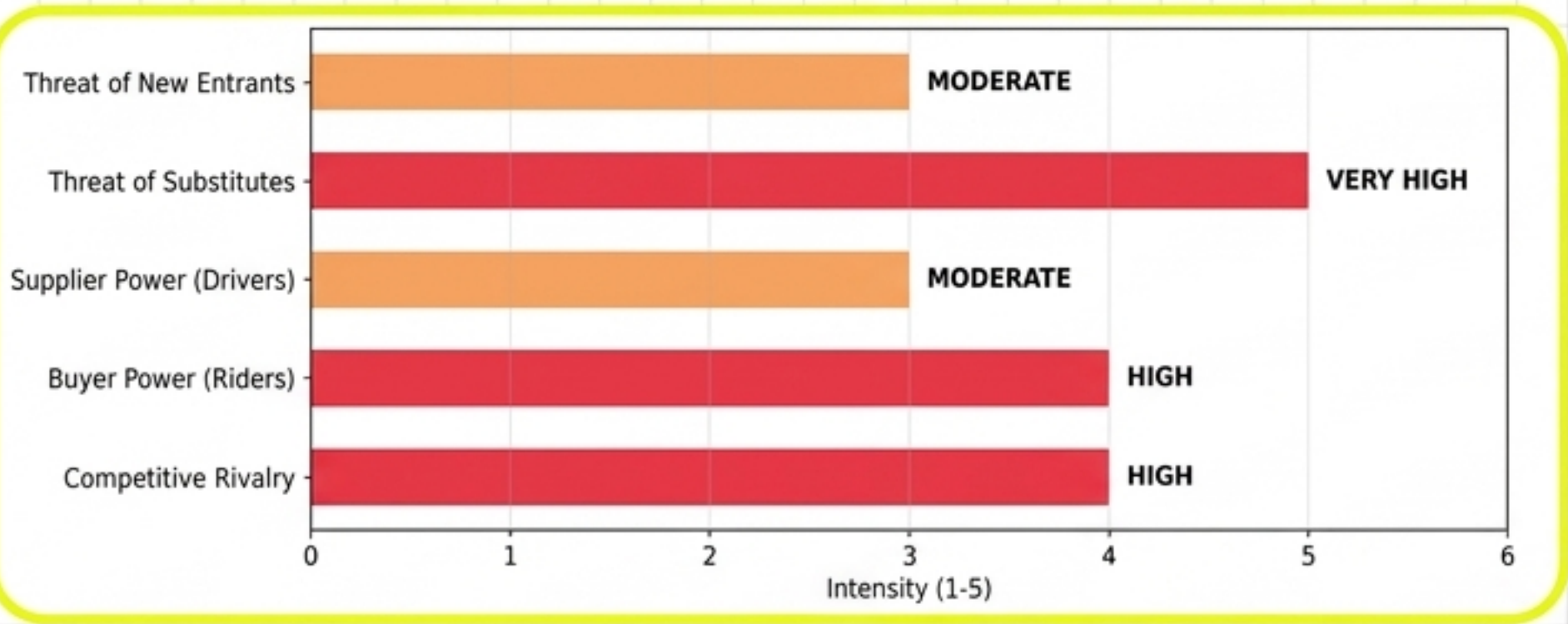
39.6%
Trips with Negative CM
99% of <3km trips at 1.0x surge

4.14 → 2.36
Satisfaction Drop (1.0x to 2.5x)
Transition begins at ~1.5x surge



Porter's Five Forces

Indian Ride-Hailing Market Structure & Pricing Defensibility Assessment



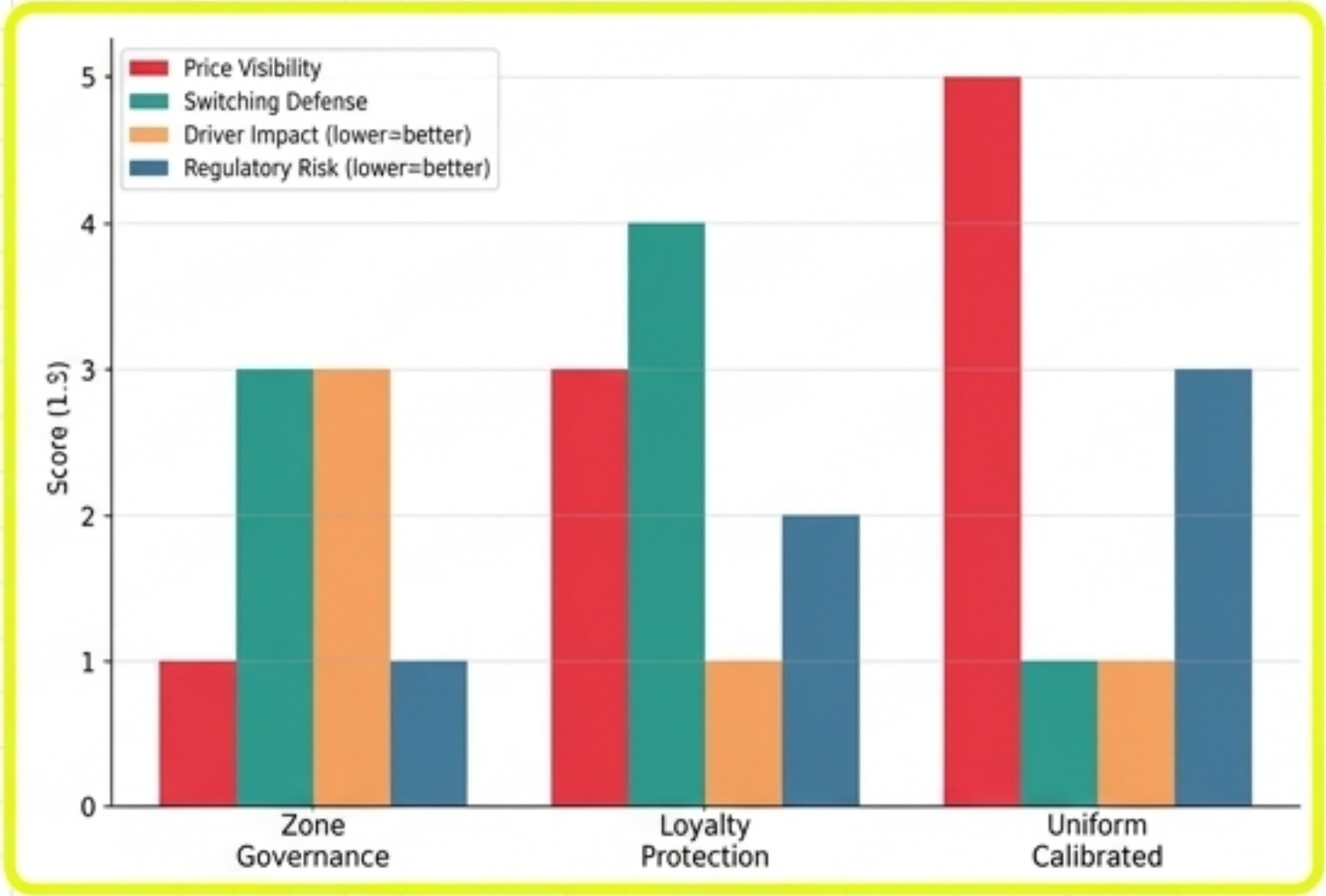
Buyer Power: HIGH
75%+ riders have 2+ ride-hailing apps installed. Multi-apping is the norm. 0% → 18% cancellation across surge.

Supplier Power (Drivers): MODERATE
Drivers multi-home but depend on platform demand. Drivers platform demand. 62.8% of CM from surge trips. Earnings multi from surge trips. Earnings guarantee is critical.

Rivalry: HIGH
3-city competitive landscape. Margin pressure from perpetual discounting. Winner-take-most dynamics.

Substitutes: VERY HIGH
Metro, autos, buses. 99% of negative-margin trips are <3km. Substitute threat ceiling is the metro fare floor.

New Entrants: MODERATE
Network effects barrier exists, but zero-surge entrants attract price-sensitive switchers.



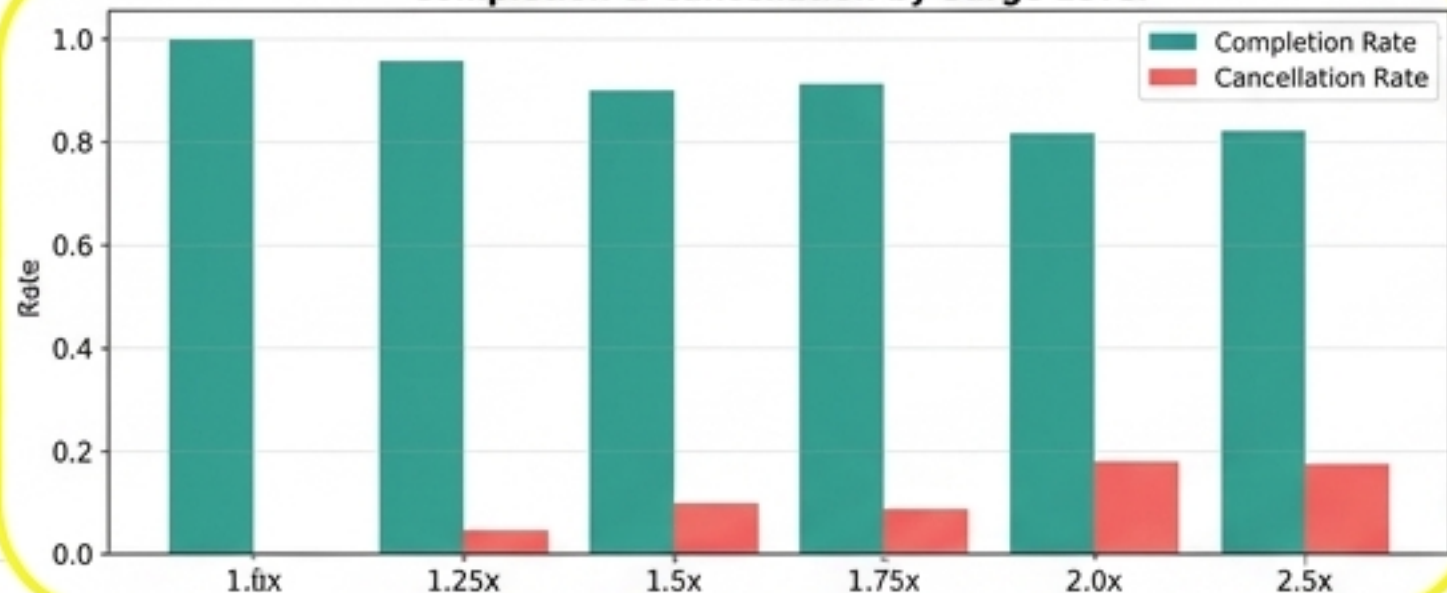
Strategy	Defensibility	Key Reason
Uniform Cap (2.0x)	LOW	Immediately observable pre-request. Competitor can match within hours.
Zone Governance	HIGH	Caps invisible until ride request. Per-zone thresholds impossible to reverse-engineer.
Loyalty Protection	MEDIUM	Tier-based. 37.1% Unknown tier lacks any protection. Hard to communicate.

KEY INSIGHT: Zone-based caps are the ONLY pricing strategy with HIGH competitive defensibility. Competitors cannot observe or replicate per-zone threshold values because caps vary by rider context... A uniform 2.0x cap is immediately observable... This information asymmetry is a durable competitive moat.

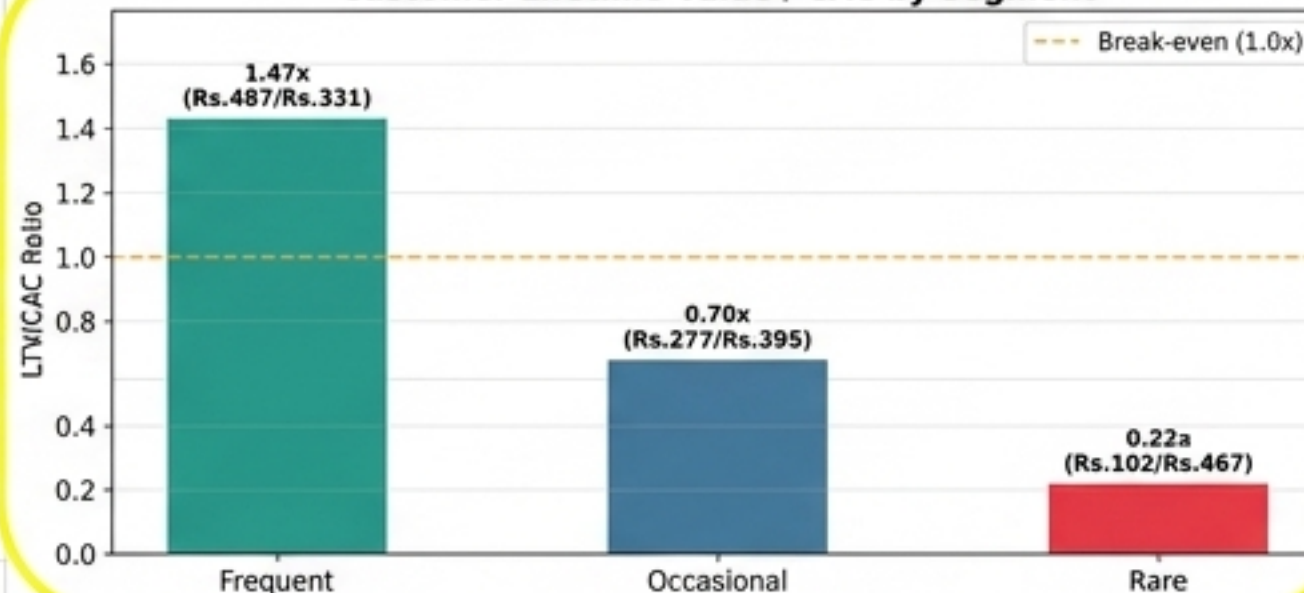
Key Analytical Findings

50,000 Trip-Level Records | 4 Workstreams | 3 Cities | 3,000 Customers

Completion & Cancellation by Surge Level



Customer Lifetime Value / CAC by Segment



Demand Suppression: Surge Erodes Trust

Satisfaction drops from **4.14 at 1.0x** to **2.36 at 2.5x surge** — a **43% decline** across the surge range. The steepest drop occurs between 1.25x and 1.5x. Cancellation rate rises from **0% at 1.0x** to **18% at 2.5x surge** — nearly 1 in 5 rides is abandoned at peak surge.

Price Elasticity: Demand is Inelastic

Arc elasticity estimated at **0.22** (97% adjusted baseline) — strongly **INELASTIC**. A 10% price increase causes only 2.2% demand loss. Validated via controlled regression with **time-of-day, zone, and weather** fixed effects. Weather confounding is **NEGLIGIBLE**.

Unit Economics: Only Frequent Riders Are Profitable

LTV/CAC by Segment (DCF, 18-month horizon, 10% annual discount): Frequent riders are **1.47x** (above breakeven). Occasional are 0.70x. Rare are 0.22x. Revenue NRR is **94.2%**.

CAVEAT: Churn model uses industry benchmarks; confidence is MEDIUM-LOW.

Negative Margin & Cross-Subsidy

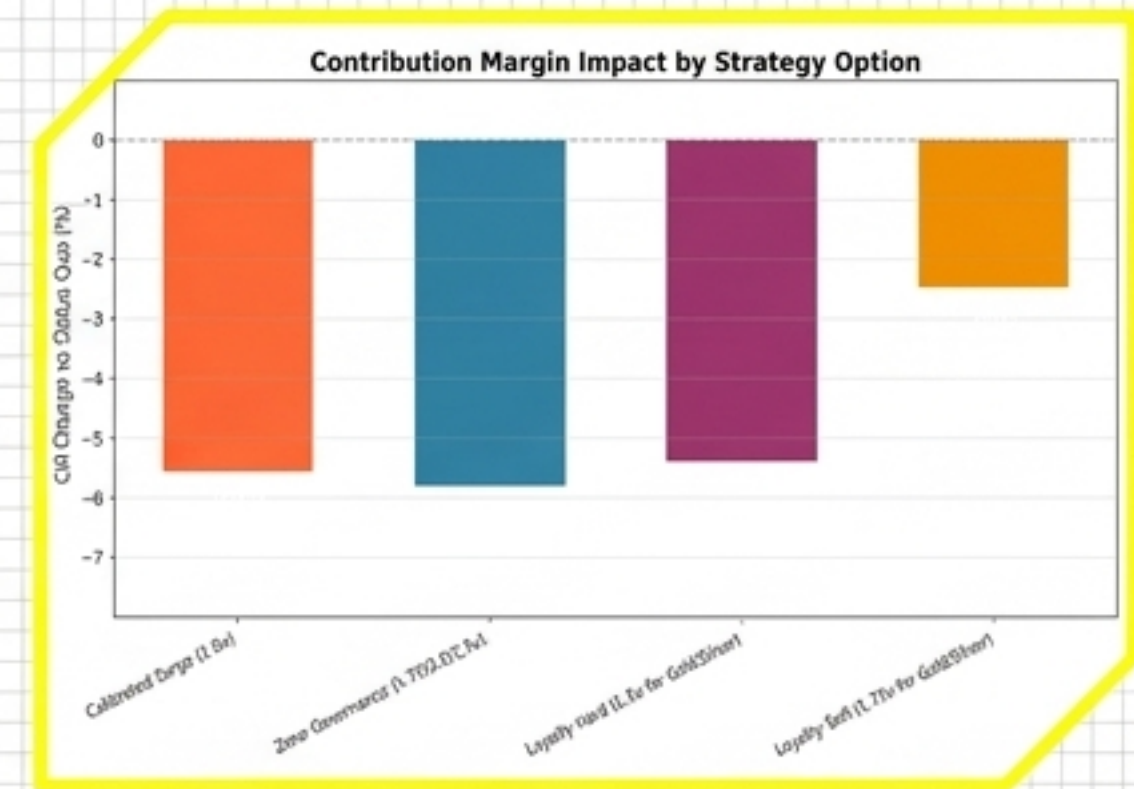
39.6% of all 50,000 trips have negative contribution margin. 99% of trips under 3km at 1.0x surge have negative CM — structurally unprofitable at current pricing. These are **NOT loss-leaders** ($r=-0.324$).

Strategic Options Comparison

Evaluating 3 Given Options + Combined Enhancement | Trip-Level Simulation (97% Baseline)

Option	Surge Caps	CM Δ	Rev Δ	Churn Δ	Cancel Δ	Trips Capped	Score
Status Quo	2.5x all zones	0.00%	0.00%	0.00pp	0.00pp	0.0%	—
S1: Calibrated Surge	2.0x uniform cap	-5.36%	-1.84%	+0.53pp	+0.05pp	10.4%	2.74
S2: Zone Gov (Recommended)	Res 1.75x / Com 2.0x / Transit 2.5x	-5.64%	-1.50%	+0.64pp	-0.57pp	10.7%	4.48
S3a: Loyalty Hard	1.5x Gold/Silver only	-5.11%	-1.42%	+0.46pp	-0.41pp	6.2%	3.03
S3b: Loyalty Soft	1.75x Gold/Silver only	-2.42%	-0.48%	+0.25pp	-0.46pp	4.7%	3.61
S4: Combined	Zone caps + 1.75x loyalty	-6.49%	-1.59%	+0.74pp	-0.84pp	12.1%	4.19

Weighted multi-criteria scoring: CM Preservation (15%), Cancellation (10%), Churn (15%), Regulatory Compliance (20%), Implementation Ease (15%), Strategic Fit (25%).



CRITICAL CAVEATS: (1) Churn magnitude is UNCERTAIN (0.15pp to 0.64pp range) — model uses benchmarks **NOT validated in this dataset**. (2) Simulation cost formula assumes fixed cost; actual CM loss likely 10-15% lower. (3) Projection-based without A/B testing. (4) Recommendation rests on a **STRUCTURAL thesis**, not churn validation.

Recommendation

Option 2: Zone Governance — Score 4.48 / 5

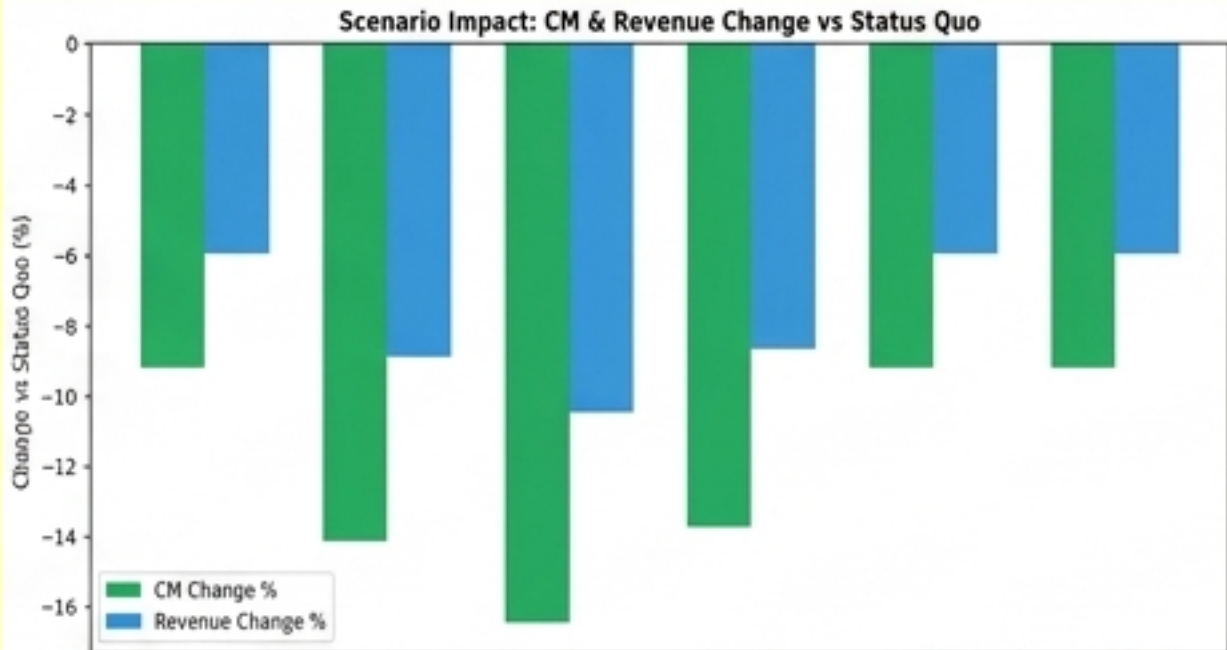
Zone-Governed Surge Pricing

Differentiated caps by zone type to preserve revenue where it matters, protect customers where substitutes abound, and position the platform as a self-regulating market leader ahead of Series D.

Residential Zones: 1.75x cap	Highest substitute availability. 41.2% negative CM rate. Steepest satisfaction drop.
Commercial Zones: 2.0x cap	Time pressure dominates price sensitivity. Moderate elasticity.
Transit Hubs: 2.5x cap	Last-mile necessity. Preserves total CM sustainability.

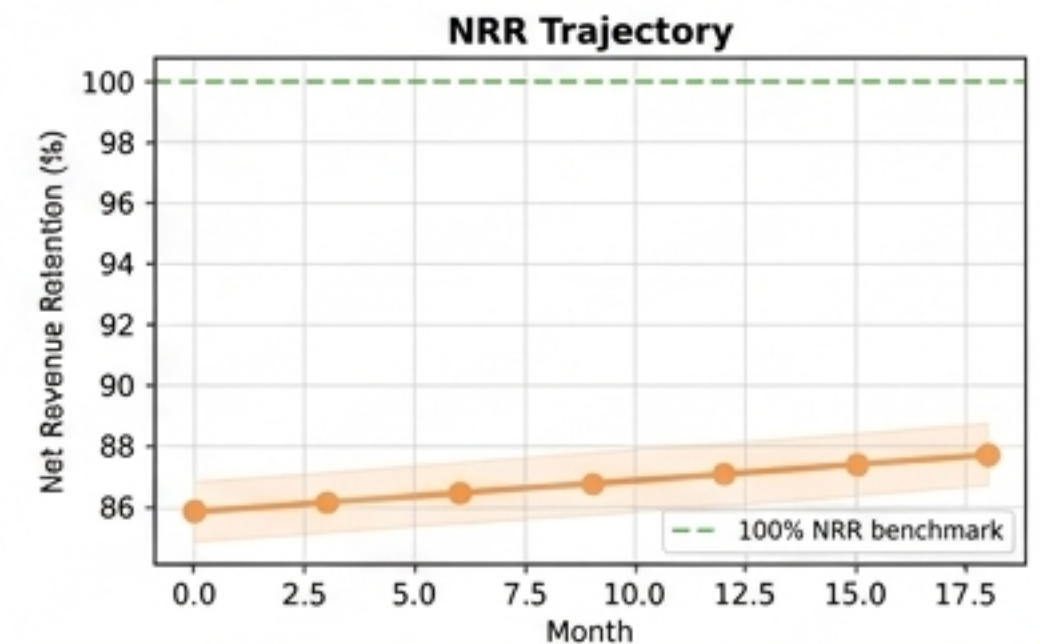
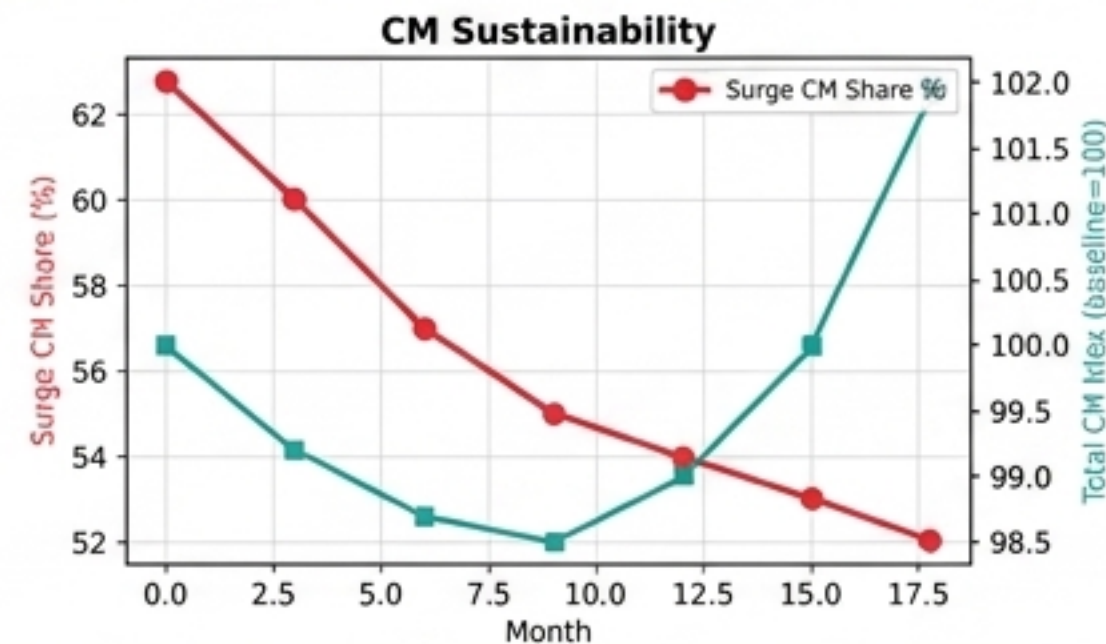
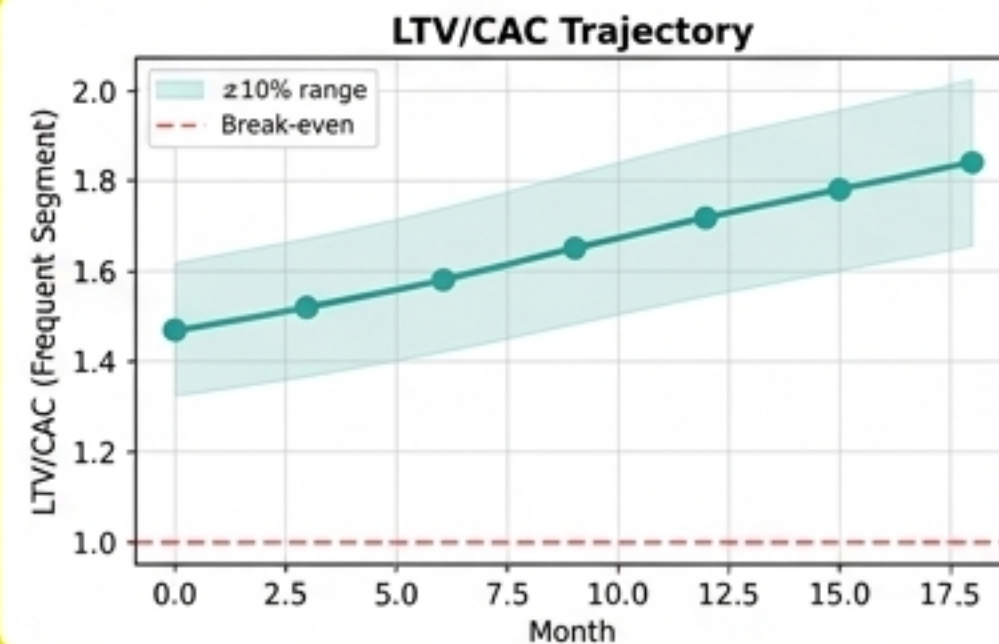
- 1. **CM Diversification:** Reduces surge dependency from 62.8% to <55%.
- 2. **Regulatory Positioning:** Preempts government-mandated uniform caps; acts as self-regulation.
- 3. **Competitive Defensibility:** Per-zone thresholds are invisible to competitors, creating a durable moat.

STATUS QUO RISK: Inaction carries 14%+ annual churn and severe regulatory vulnerability. The **cost of inaction exceeds the cost of action.**



Investor Narrative

Three Metrics, Three Directional Statements, 18-Month Horizon to Series D



LTV/CAC Ratio (Frequent Segment)

1.47x → 1.47-1.84x

Driver: Churn reduction and satisfaction improvement.

Confidence: ●●○ **MEDIUM**

Presented strictly as a RANGE due to churn model uncertainty.

CM Sustainability

62.8% → <55%

Driver: Minimum fare rules and volume-driven composition.

Confidence: ●●● **MEDIUM-HIGH**

Structurally robust; elasticity sensitivity is strictly bounded.

Net Revenue Retention (NRR)

94.2% → ≥95%

Driver: Residential satisfaction up, churned customers represent low revenue.

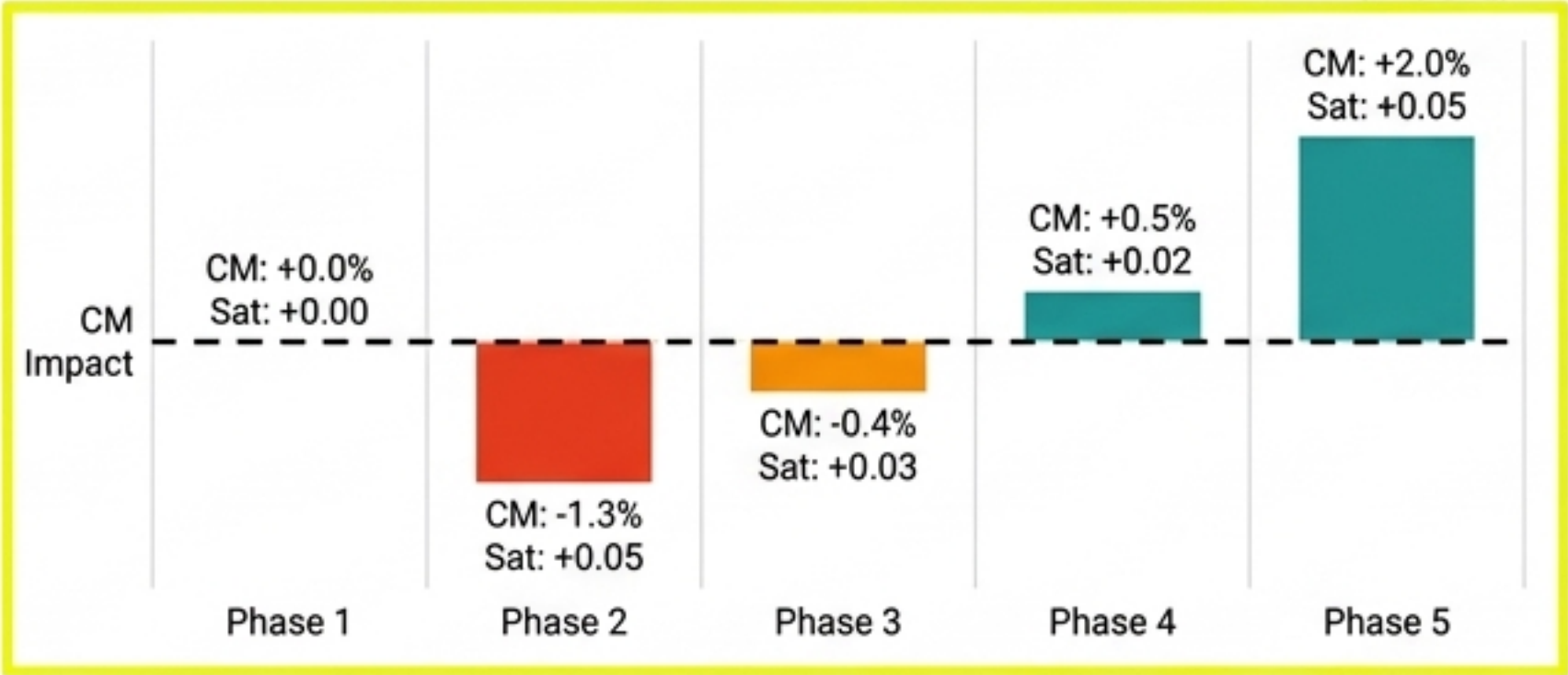
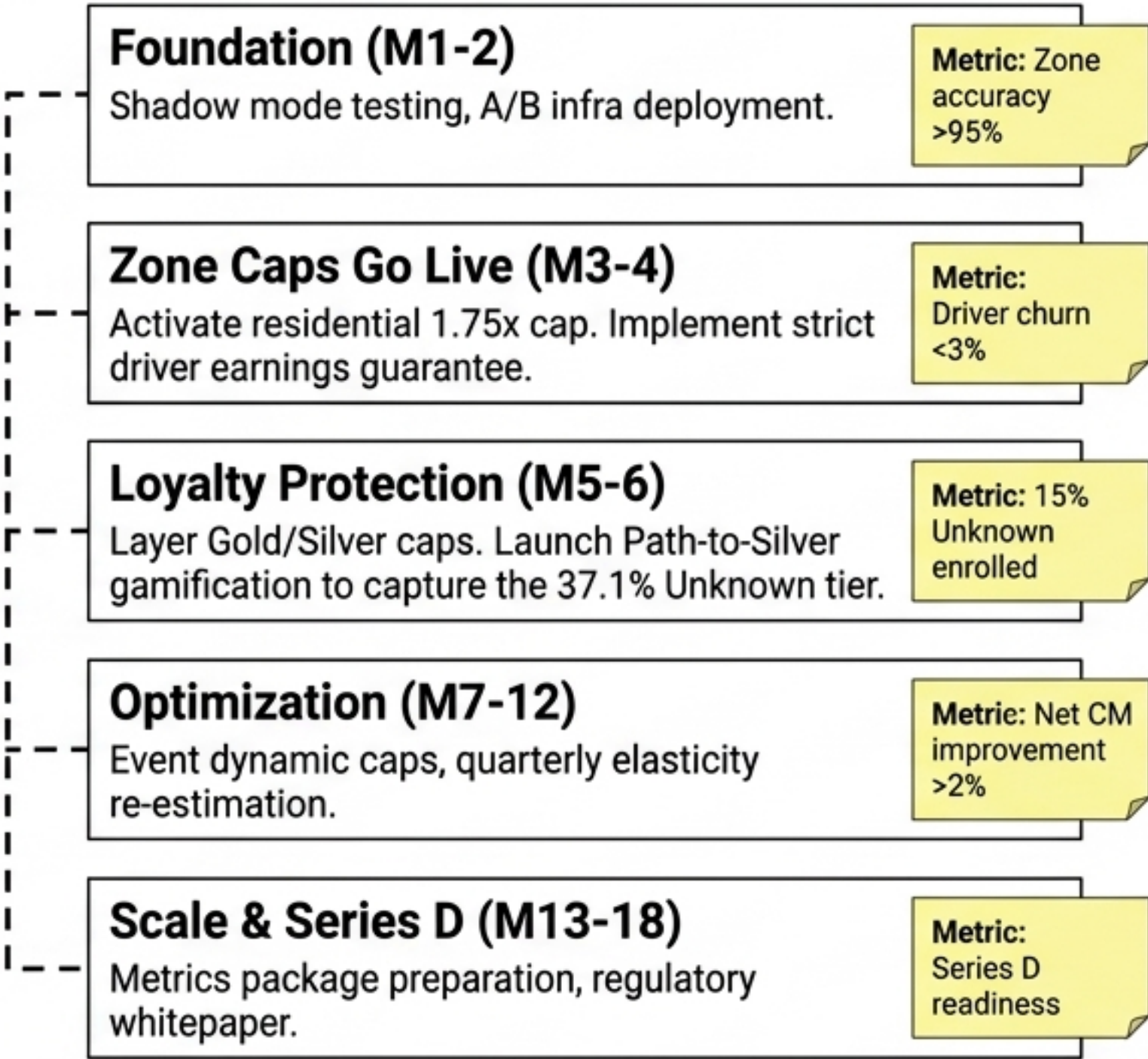
Confidence: ●●● **MEDIUM-HIGH**

NRR is already excellent by SaaS standards.

The 18-month trajectory is **DIRECTIONALLY SOUND** but **MAGNITUDE UNCERTAIN**. The structural thesis does NOT depend on churn model validation. The recommendation is commercially viable even at the pessimistic lower bound of all churn assumptions.

Implementation Roadmap

5-Phase, 18-Month Rollout | 1 Critical Risk (Driver Earnings) | Rs.8-12 Cr Budget



Severity	Risk	Mitigation
CRITICAL	Driver earnings reduction in residential zones	6-month earnings guarantee floor (90%). Strict budget: Rs. 5Cr.
HIGH	Competitor matching	Leverage zone-based opacity. Invisible thresholds.
HIGH	Unknown tier alienation	Path-to-Silver gamified tier progression.
MEDIUM	Regulatory intervention	Publish self-regulation whitepaper.

Appendix

Methodology, Data Quality Caveats, Sensitivity Analysis & Chart References

A. Analytical Methodology Summary

WS1: Demand Suppression (Group-by analysis, loss-leader rejected $r=-0.324$).

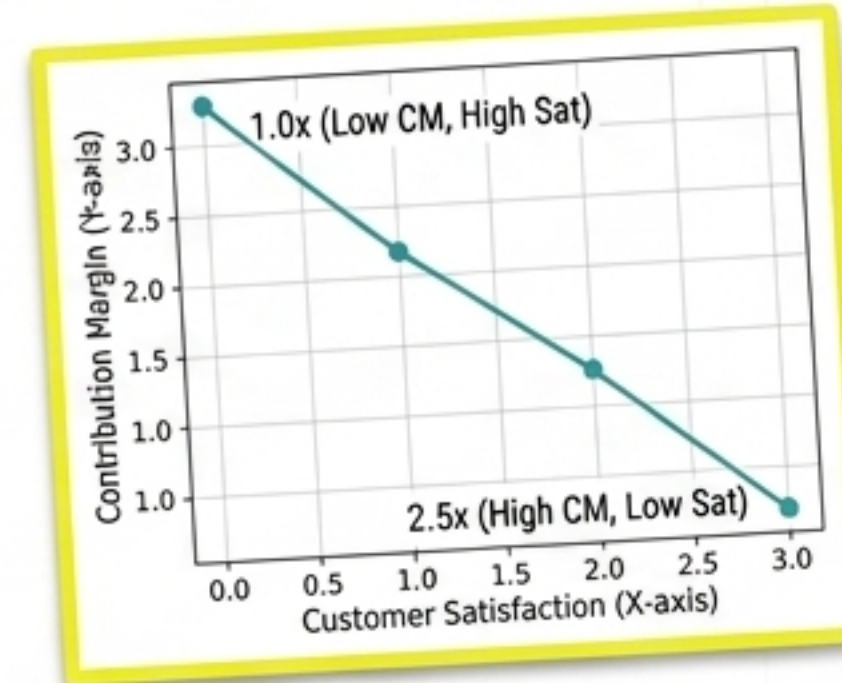
WS2: Price Elasticity (Arc elasticity formula, 0.22 validated via controlled regression).

WS3: Unit Economics (DCF LTV 18-month, 10% discount, bracketed churn binary search).

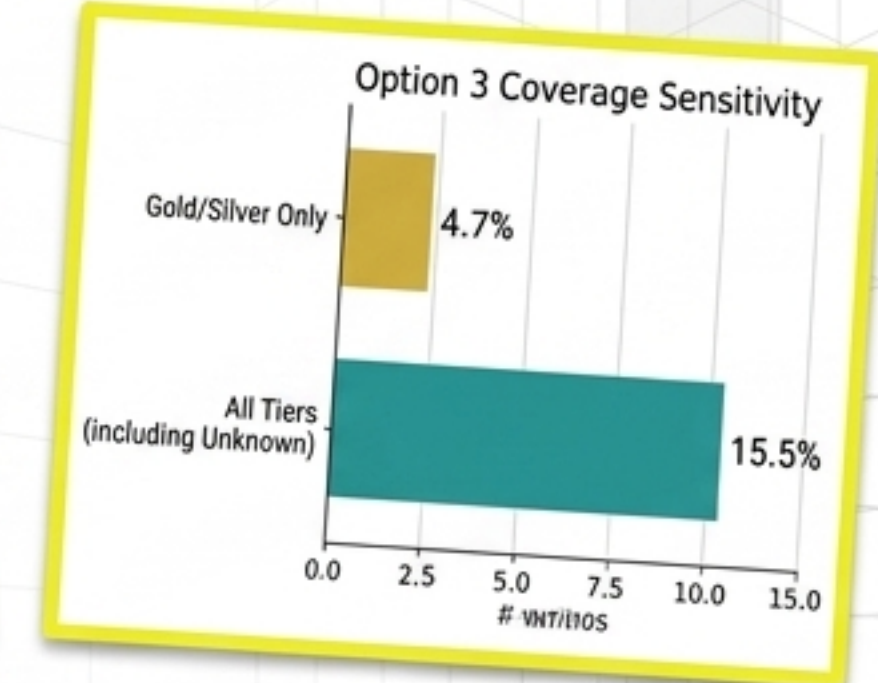
WS4: Scenario Simulation (Trip-level, conservative fixed cost assumption).

B. Known Data Limitations

1. 100% completion at 1.0x surge is a raw artifact; manually adjusted to 97% baseline.
2. Non-monotonic completion at 1.75x artifact (interpolated).
3. 37.1% Unknown loyalty tier gap.
4. Churn causality is NOT proven (behavioral correlations ~ 0.0). Model is unvalidated.
5. Simulation is projection-based; A/B testing is required.



*CM vs Satisfaction Trade-off:
The core tension of surge pricing.*



*37.1% Unknown tier represents
a massive protection gap.*

C. Scoring Robustness to Weight Assumptions

Zone Governance scores across weight configurations: Base (4.48), CM-Heavy (4.16), Churn-Heavy (4.55), Reg-Heavy (4.51), Strat-Heavy (4.60), Equal (4.27). Zone Governance wins under ALL 6 permuted scenarios and NEVER loses to the Combined strategy. Recommendation is mathematically robust.